4 (a)

$$\omega^2 = \frac{k}{m} \Rightarrow \left(\frac{2\pi}{T}\right)^2 = \frac{k}{m}$$

$$T^2 = \frac{4\pi^2 m}{k} \Rightarrow T \propto \sqrt{\frac{m}{k}}$$ C1

$$\frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1} \times \frac{k_1}{k_2}} = \sqrt{\frac{2m}{m} \times \frac{k}{0.5k}} = \sqrt{4} = 2$$ A1

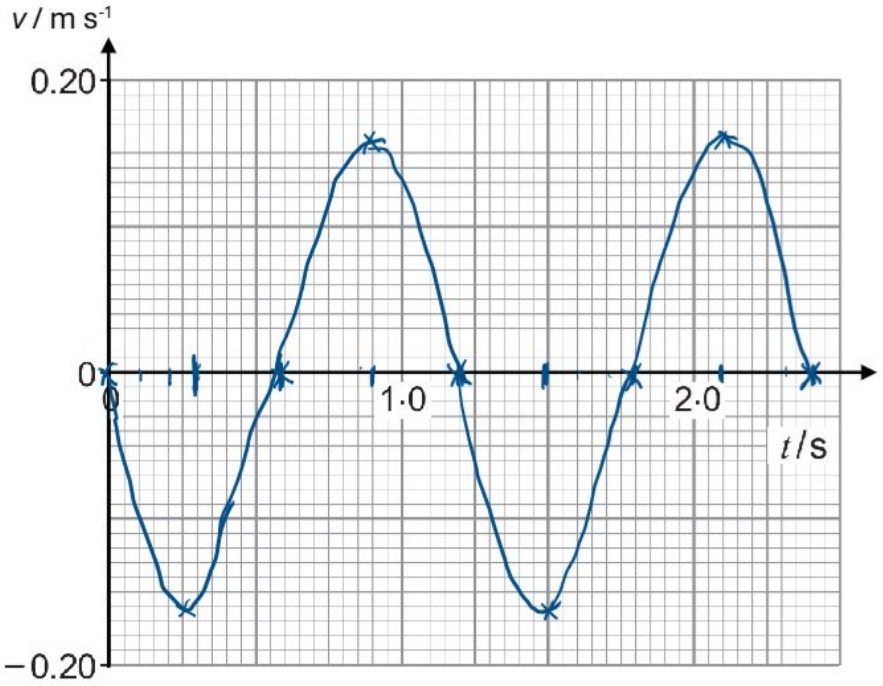
(b)(i)

$$v_{\max} = \omega x_o = \left(\frac{2\pi}{T}\right) x_o$$ M1

$$= \left(\frac{2\pi}{1.20}\right)\left(30.0 \times 10^{-3}\right)$$

$$= 0.157 = 0.16 \text{ m s}^{-1}$$ A0

(b)(ii)



inverted sine graph B1
period 1.2 s, 2 oscillations drawn, peak value 0.16 m s^{-1} B1

(b)(iii)

$$v = -0.157 \sin \omega t = -0.157 \sin\left(\frac{2\pi}{1.20}\right)(3.50)$$ C1

$$= 0.079 \text{ m s}^{-1}$$ A1

(Alternative: Speed the same as at 1.1 s and 2.3 s. Speed = 0.079 m s^{-1})

(b)(iv) T = 1/1.20 = 0.833 Hz, 1.67 Hz is twice the oscillation frequency. B1
This is correct because KE = ½ mv^2 and v^2 will have two [equal] positive peaks per oscillation cycle. B1

5 (a)(i) When one polaroid is rotated, resultant intensity of bright fringes decreases but the B1